JAY ADESARA | ME23B002 | INDIAN INSTITUTE OF TECHNOLOGY MADRAS

GitHub | LinkedIn | Codeforces

EDUCATION

Program	Institute	% / CGPA	Year of Completion
B.Tech in Mechanical Engineering Minor in Computing*	Indian Institute of Technology, Madras	9.06	2027
Class XII (GSEB)	B.H. Rathod Sec. and Higher Sec. School, Rajkot	87.38	2023
Class X (GSEB)	Chanakya Vidhya Mandir Sankul, Rajkot	99.16	2021

Achievements

- Secured **Top 75** in **Code with Cisco 2026**, earning selection to Cisco's Bangalore campus for **Code - A - Thon**, where I built an autonomous observability platform performing causal root-cause analysis for microservices.
- Ranked **24th** in **InMobi Codies 2026** among 12k students; **Top 10** in **DaVinci Trading League** among 400 students
- Awarded **Reliance Foundation Undergraduate Scholarship** among 58,000+ applicants through an aptitude test
- Secured state rank **1** in **class X** with **99.99** Percentile and state rank **149** in **ACPC** counselling among 136k+ candidates.

PROFESSIONAL EXPERIENCE

Quantitative Brokers
SDE Internship
May' 26 - July' 26

Built OpenTelemetry **observability** POC to replace fragmented log-based debugging with unified request-level **RCA**

- Designed and built a POC observability pipeline for Node.js microservices using **OpenTelemetry**, routing traces, metrics, & logs to **Jaeger**, **Prometheus**, & **Graylog**, visualized in **Grafana**, reducing debugging time by **60%**
- Demonstrated request-level **root-cause analysis** by correlating metrics to traces via **Prometheus exemplars** and logs to traces via Pino **trace-ID** injection, pinpointing failures to the exact service, function, and time per hop.
- Implemented span-derived **RED metrics** via the OpenTelemetry Collector's **spanmetrics connector**, bypassing JavaScript's limited exemplar support, and unified service health, latency, errors, and traces in Grafana.

WorldQuant BRAIN
Research Consultant
May' 25 - Jul' 25

- Developed and optimized **20+ quantitative models** to forecast the trends in PnL across global financial markets
- Applied robust statistical techniques to refine models and to optimize Sharpe ratio, covariance, and correlation
- Designed and back tested **long-short market-neutral** alphas to minimize volatility and enhance performance

PROJECTS

PageMaster
[Memory Unit Simulation]

Developed a **C/C++ memory management** engine simulating MMU with **paging**, sustaining 7800 operations/sec

- Implemented **LRU page replacement** with swap - in/swap - out logic for efficient frame reuse under pressure
- Engineered **dynamic page-table management** per process, with bounds checks for invalid & overflow addresses
- Built a **command-driven interpreter** for process execution, memory inspection, & live page-table introspection
- Benchmarked across **25 runs**, confirming stable LRU eviction ordering at **7,800 ops/sec** under memory pressure

Trident
[Multi - Client Chat Server]

Engineered a multi-client chat server in **C** supporting **fork**-based, **thread**-based, and **select**-based server models

- Implemented **broadcast**, **private messaging**, and **user authentication** identically across all three server models
- Designed a custom newline-framed protocol with a dedicated discovery server for user registration and lookup
- Built Python benchmarking suite to profile **latency**, **throughput**, **CPU usage**, & **memory** across concurrent clients
- Diagnosed head-of-line blocking in the broadcast loop as the shared bottleneck behind identical server latency

AlphaForge
[Market Data & Strategy Analytics Platform]

Built AlphaForge, a market research platform ingesting historical **OHLCV** data via the **yfinance** package for analysis

- Rendered interactive React **dashboards** with price charts and technical indicators to evaluate trading strategies
- Designed a custom strategy-expression engine enabling users to define and **backtest** signals via a lightweight **DSL**
- Built an automated trade simulation engine computing **P&L**, **Sharpe** ratio, and maximum **drawdown** per strategy
- Architected the platform with a React frontend and Node.js/Express backend serving strategy and market data

StratNet4
[Connect 4 Strategy Arena]

Built StratNet4, a Connect 4 strategy arena comparing **Random, Minimax, Alpha - Beta, and MCTS** playing agents

- Implemented four AI agents and ran round - robin tournaments to compare relative playing strength across all
- Measured search efficiency per strategy to analyze the **trade-off** between compute cost & the playing strength
- Evaluated **pruning** and rollout - based strategies to quantify accuracy and speed trade-offs under time budgets

B.Tech Project*
Prof. Kameswarao Anupindi

Immersed Boundary Method in an SLBM Framework for Porous Media & Moving-Body Flows | Aug' 26 – Ongoing*

- Implementing an immersed boundary method in an **SLBM** framework to study porous media flows and moving bodies with imposed kinematics, while optimizing computational performance via **OpenMP** & **MPI** parallelization

RELEVANT SKILLS

Languages And Libraries	C/C++, Python, OpenMP, MPI, OpenACC, MATLAB, Git, GitHub, SQL, Power BI React, HTML, CSS, JavaScript, Node.js, Jaeger, Prometheus, Graylog, Grafana
Coding Experience	Achieved **Specialist** title on **Codeforces** with peak rating of **1409**, showcasing competitive programming progress Secured global rank of **1014** in Codeforces Round 1031 (Division 2) among the 19,000+ participated candidates

LEADERSHIP ENDEAVORS & EXTRA-CURRICULARS

Core Member of Shaastra*
OnIP

- Nominated by the Dean of Students to be part of the exclusive **24 - member core organizing team** of Shaastra
- Spearheading a 3-tiered team of **60+** students and a budget of over **INR 15M** to plan and run the festival

Head
GGS

- Led a team of **20+** members across **2** consecutive years to organize IIT Madras's flagship Navratri festival, planning and executing multiple events for **1000+** participants while overseeing an annual budget of **₹ 2 lakh**

CFI

Team Anveshak

- Engineered a 3D-printed TPU wheel with W-shaped tread and hexagonal infill for **16% weight reduction**, and designed direct-drive independent steering system with in-house fork housing, cutting sub-system weight by **4 kg**

Mathematics Club

- Learned Game Theory fundamentals - Nim games, zero-sum games, & N/P positions, through active participation

Trading

- Managing a diversified **portfolio** of **INR 1,00,000+** of equity and mutual funds with the maximum return of **22.5%**

*- Ongoing | GSEB - Gujarat Secondary and Higher Secondary Board | ACPC - Admission Committee for Professional Courses | SDE - Software Development Engineer | POC - Proof of Concept | RCA – Root Cause Analysis | RED - Rate, Error and Duration | PnL - Profit and Loss | MMU - Memory Management Unit | LRU - Least Recently Used | OHLCV - Open, High, Low, Close and Volume | DSL - Domain-Specific Language | MCTS - Monte Carlo Tree Search | SLBM - Simplified Lattice Boltzmann Method | INR - Indian Rupees | TPU - Thermoplastic Polyurethane | CFI - Center For Innovation | OnIP - Operations and Infrastructure Planning | GGS - Garvi Gujarat Sangathan